Chapter 12: Distributed Database Management Systems

Section A: Multiple Choice

1. A distributed database management system (DDBMS) governs the storage and processing of logically related data over:
   a) A single centralized site
   b) Interconnected computer systems where data and processing are distributed among several sites
   c) A single mainframe computer
   d) Dumb terminals connected to a host

2. In a distributed processing environment, the database's logical processing is shared among two or more physically independent sites connected through:
   a) A single CPU
   b) A network
   c) A dumb terminal
   d) A local disk

3. A distributed database stores a logically related database over:
   a) One physical site
   b) Two or more physically independent sites
   c) A single file server
   d) A single mainframe

4. Which of the following is an advantage of a DDBMS?
   a) Complexity of management and control
   b) Lack of standards
   c) Faster data access
   d) Increased training cost

5. Which of the following is a disadvantage of a DDBMS?
   a) Faster data processing
   b) Security
   c) Reduced operating costs
   d) Processor independence

6. The software component found in each computer that requests data in a DDBMS is called the:
   a) Data processor
   b) Transaction processor
   c) Lock manager
   d) Scheduler

7. In a fully distributed DBMS, the TP is also known as the:
   a) Data manager
   b) Application processor
   c) Lock manager
   d) Data processor

8. A DDBMS that integrates only one type of centralized DBMS over a network is called:
   a) Heterogeneous
   b) Homogeneous
   c) Mixed
   d) Fragmented

9. A DDBMS that integrates multiple instances of different DBMSs over a network is called:
   a) Homogeneous
   b) Heterogeneous
   c) Centralized
   d) Replicated

10. A DDBMS that integrates multiple instances of different DBMSs over a network is called:
    a) Homogeneous
    b) Heterogeneous
    c) Centralized
    d) Replicated

11. Which transparency feature allows a distributed database to be treated as a single logical database?
    a) Transaction transparency
    b) Failure transparency
    c) Distribution transparency
    d) Performance transparency

12. Which transparency feature allows a transaction to update data at more than one network site?
    a) Distribution transparency
    b) Transaction transparency
    c) Failure transparency
    d) Heterogeneity transparency

13. Which transparency feature ensures that the system will continue to operate in the event of a node failure?
    a) Performance transparency
    b) Distribution transparency
    c) Failure transparency
    d) Transaction transparency

14. Which transparency feature allows the integration of several different local DBMSs under a common, or global, schema?
    a) Heterogeneity transparency
    b) Distribution transparency
    c) Transaction transparency
    d) Failure transparency

15. A single SQL statement that accesses data to be processed by a single remote database processor is called a:
    a) Distributed request
    b) Remote request
    c) Distributed transaction
    d) Remote transaction

16. A transaction composed of several requests that accesses data at a single remote site is called a:
    a) Remote request
    b) Distributed request
    c) Remote transaction
    d) Distributed transaction

17. A transaction that can reference several different local or remote DP sites is called a:
    a) Remote transaction
    b) Distributed transaction
    c) Remote request
    d) Distributed request

18. A single SQL statement that references data located at several different local or remote DP sites is called a:
    a) Remote request
    b) Distributed request
    c) Remote transaction
    d) Distributed transaction

19. The two-phase commit protocol (2PC) guarantees that:
    a) All transactions are committed immediately
    b) If a portion of a transaction cannot be committed, all changes at other sites will be undone
    c) Only one site commits its part of the transaction
    d) Transactions are never rolled back

20. In a DDBMS, concurrency control is important because:
    a) Multisite, multiple-process operations are more likely to create data inconsistencies and deadlocked transactions
    b) Single-site systems are more complex
    c) Data is stored at only one site
    d) Transactions are never concurrent

Section B: True/False

1. A distributed database stores logically related data in two or more physically independent sites connected via a computer network.
2. In a distributed processing system, the database is composed of several parts known as database fragments.
3. A homogeneous DDBMS integrates multiple instances of different DBMSs over a network.
4. Distribution transparency allows a distributed database to be treated as a single logical database.
5. Transaction transparency allows a transaction to update data at more than one network site.
6. Failure transparency ensures that the system will continue to operate in the event of a node failure.
7. Heterogeneity transparency allows the integration of several different local DBMSs under a common, or global, schema.
8. Fragmentation transparency is the lowest level of distribution transparency.
9. Location transparency exists when the end user must specify both the fragment names and their locations.
10. A remote request lets a single SQL statement access data to be processed by a single remote database processor.
11. A distributed transaction can reference several different local or remote DP sites.
12. A distributed request lets a single SQL statement reference data located at several different local or remote DP sites.
13. The two-phase commit protocol guarantees that if a portion of a transaction cannot be committed, all changes made at other sites will be undone.
14. The DO-UNDO-REDO protocol is used by the DP to roll transactions back and forward with the help of transaction log entries.
15. The write-ahead protocol forces the log entry to be written to permanent storage after the actual operation takes place.
16. Horizontal fragmentation divides a relation into attribute (column) subsets.
17. Vertical fragmentation divides a relation into subsets of tuples (rows).
18. Mixed fragmentation is a combination of horizontal and vertical fragmentation strategies.
19. Data replication can enhance data availability and response time.
20. A fully replicated database stores multiple copies of each database fragment at multiple sites.
21. A partially replicated database stores each database fragment at a single site.
22. An unreplicated database stores each database fragment at a single site with no duplicate fragments.
23. Centralized data allocation stores the entire database at one site.
24. Partitioned data allocation divides the database into several disjointed parts and stores them at several sites.
25. The transaction processor (TP) is also known as the data manager (DM).
26. The data processor (DP) is also known as the application processor (AP).

27. A DDBMS must perform all the functions of a centralized DBMS.
28. Performance transparency allows a DDBMS to perform as if it were a centralized database.
29. Query optimization in a DDBMS only considers CPU time cost.
30. The design of a distributed database must consider fragmentation, replication, and allocation.
31. A distributed database system requires distributed processing.
32. The main components of a DDBMS are the transaction processor and the data processor.
33. In a DDBMS, each TP can access data on any DP, and each DP handles all requests for local data from any TP.
34. The two-phase commit protocol requires a DO-UNDO-REDO protocol and a write-ahead protocol.
35. The objective of query optimization is to minimize the total cost associated with the execution of a request.

Section C: Matching Columns

Column A	Column B
1. DDBMS	a) A single SQL statement accesses data processed by a single remote database processor
2. Distributed database	b) Governs storage and processing of logically related data over interconnected computer systems
3. Transaction processor (TP)	c) Stores a logically related database over two or more physically independent sites
4. Data processor (DP)	d) Software component found in each computer that requests data
5. Distribution transparency	e) Software component residing on each computer that stores and retrieves data located at the site
6. Transaction transparency	f) Allows a distributed database to be treated as a single logical database
7. Fragmentation transparency	g) Allows a transaction to update data at more than one network site
8. Remote request	h) Highest level of transparency; end user does not need to know database is partitioned
9. Two-phase commit	i) Guarantees that if a portion of a transaction cannot be committed, all changes at other sites will be undone
10. Horizontal fragmentation	j) Division of a relation into subsets of tuples (rows)

Section D: Short Questions

1. Explain the purpose and function of a distributed database management system (DDBMS).
2. Give the advantages and disadvantages of DDBMSs.
3. Describe the characteristics and components of a DDBMS.
4. Explain how database implementation is affected by different levels of data and process distribution.
5. Describe how transactions are managed in a distributed database environment.
6. Differentiate between distributed processing and distributed databases.
7. Describe the DDBMS components and explain the role of each component.
8. Explain the evolution of distributed database management systems and the factors that drove their development.
9. Differentiate between fully replicated, partially replicated, and unreplicated databases.
10. Describe the three types of data fragmentation strategies: horizontal, vertical, and mixed fragmentation.
11. Explain the two-phase commit protocol (2PC) and why it is necessary in a distributed database environment.
12. Differentiate between a remote request, a remote transaction, a distributed transaction, and a distributed request.